

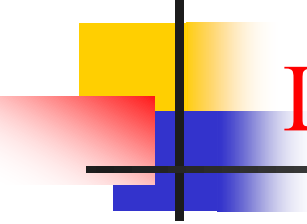
Computer & Internet Ethics

Digital Ethics: Challenges and Solutions

Dr. Ghaith Saker

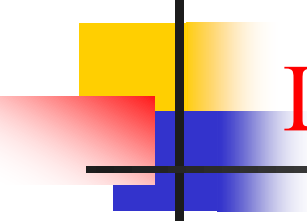
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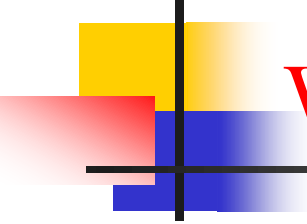
Lecture Outline

- Introduction to Digital Ethics
- Key Ethical Dimensions
 - Privacy and Surveillance
 - Cybercrime and Fraud
 - Intellectual Property
 - Information Quality
 - The Digital Divide and Algorithmic Bias
 - Censorship and Freedom of Expression
 - Accountability and System Quality



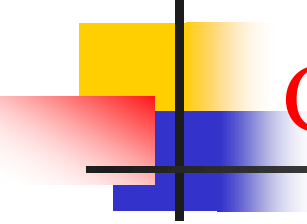
Lecture Outline

- ❑ Artificial Intelligence and Ethics
- ❑ Open Educational Resources
- ❑ Ethical Hacking
- ❑ Digital Literacy
- ❑ Shared Responsibility Framework
- ❑ Recommendations and Conclusion



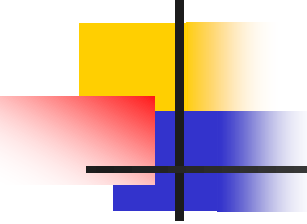
What is Digital Ethics?

- ❖ **Comprehensive Definition:** The principles and standards that govern responsible behavior in the digital world.
- ❖ The study of ethical issues arising from the use of technology.
- ❖ Applying traditional ethical values to digital contexts.



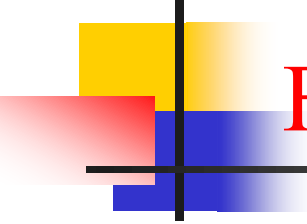
Core Principles of Digital Ethics

- ❖ Mutual Respect
- ❖ Social Responsibility
- ❖ Transparency and Clarity
- ❖ Justice and Fairness
- ❖ Privacy Protection
- ❖ Accountability and Responsibility



Ethical Dimensions

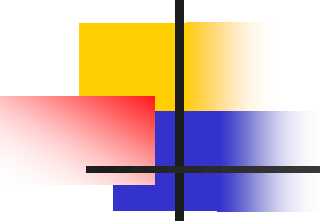
- ❖ Privacy and Surveillance
- ❖ Cybercrime and Fraud
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First Dimension: Privacy and Surveillance

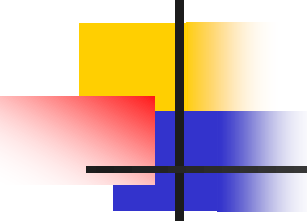
- ❖ **Privacy Risks:**

- Cookies: Tracking digital activity.
- Spyware: Covertly collecting information.
- Employee Monitoring: Email and activity.
- Keystroke Loggers: Recording keystrokes.



Privacy Protection Solutions

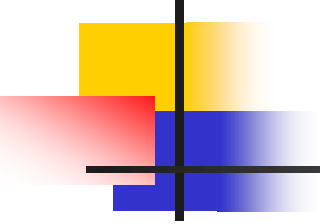
- Clear privacy policies.
- Data protection legislation.
- Anti-tracking tools.
- Encryption and security tools.
- Awareness of digital risks



Second Dimension: Cybercrime and Fraud

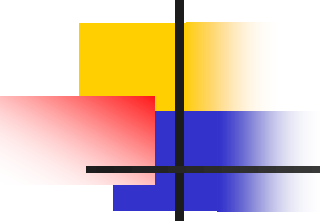
❖ **Forms of Cybercrime:**

- Phishing Attacks
- Denial-of-Service (DoS) Attacks
- Ransomware
- Identity Theft
- Impersonation



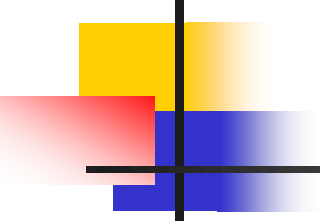
Shocking Global Statistics

- Cybercrime costs trillions of dollars annually.
- 300% increase in cyberattacks since 2020.
- 73% of organizations experienced a data breach in 2024.
- 95% of security breaches are due to human error.



Identity Theft - Devastating Effects

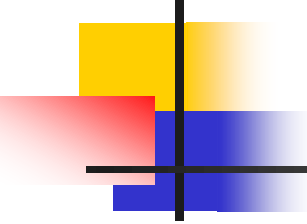
- Destruction of credit rating.
- Unexpected legal problems.
- Significant financial losses.
- Damage to victims' reputation.
- Difficulty in restoring a stolen identity.



Third Dimension: Intellectual Property

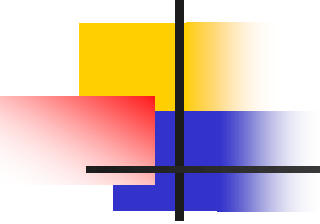
❖ **Types of Protection:**

- Copyright
- Patents
- Trademarks
- Trade Secrets



Intellectual Property Challenges

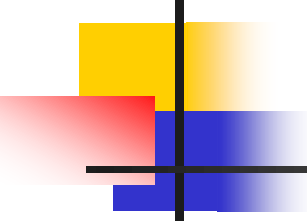
- Easy digital copying.
- Difficulties in global enforcement.
- Conflicting international laws.
- Challenges in identifying violations.
- Difficulty in tracking and monitoring.



Fair Use Principle

❖ **Conditions for Fair Use:**

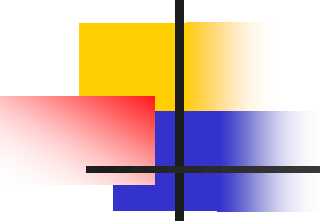
- Educational purposes.
- Scientific research.
- Criticism and review.
- News reporting.
- Parody



Fourth Dimension: Information Quality

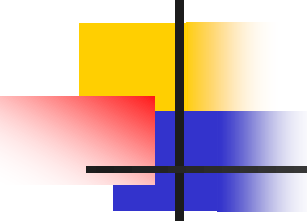
❖ **Information Quality Problems:**

- Fake News
- Conspiracy Theories
- Inaccurate Information
- Media Bias
- Deliberate Misinformation



Impact of Misinformation

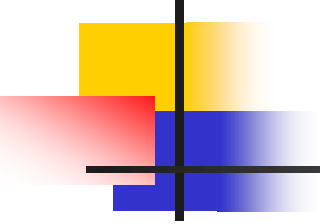
- Manipulation of public opinion.
- Election interference.
- Risks to public health.
- Incitement of social conflicts.
- Undermining trust in institutions



Filter Bubbles and Echo Chambers

❖ **How Filter Bubbles Work:**

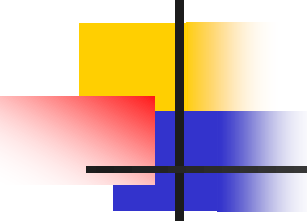
- Algorithms show content that matches your interests.
- Isolation from opposing viewpoints.
- Reinforcement of existing biases.
- Creation of a distorted digital reality.



Fifth Dimension: The Digital Divide

- ❖ **Levels of the Digital Divide:**

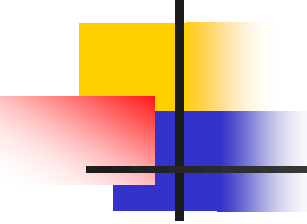
- Access: Availability of infrastructure.
- Skills: Ability to use effectively.
- Content: Availability of relevant content.
- Benefit: Transition from usage to tangible benefit



Sixth Dimension: Algorithmic Bias

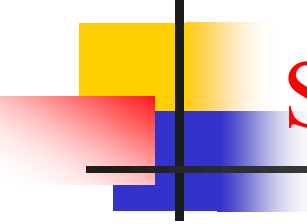
❖ **Forms of Algorithmic Bias:**

- Racial Discrimination
- Gender Bias
- Age Discrimination
- Economic Bias



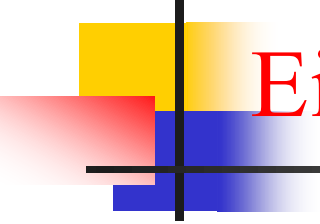
Real-World Examples of Algorithmic Bias

- Biased hiring systems.
- Unfair lending algorithms.
- Facial recognition less accurate for some ethnicities.
- Biased service recommendations



Seventh Dimension: Accountability and System Quality

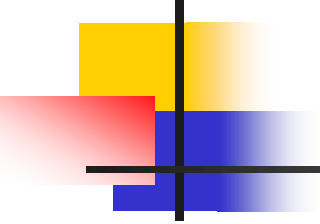
- ❖ **Accountability Challenges:**
 - Difficulty determining responsibility for system errors.
 - Complex chains of responsibility.
 - Difficulty interpreting AI decisions.
 - Conflicting international laws



Eighth Dimension: Censorship and Freedom of Expression

❖ **Global Censorship Models:**

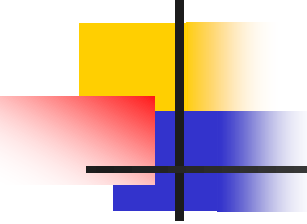
- Chinese Model: Strict government censorship.
- European Model: Data protection and freedoms.
- American Model: Freedom of speech with limits.
- Islamic Model: Moral and religious censorship



Open Educational Resources (OER)

❖ **Advantages of OER:**

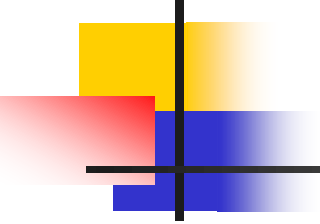
- Free Access.
- Ability to Modify.
- Freedom to Share.
- Collaborative Development.
- Full Transparency.



Ethical Hacking

- ❖ **Goals of Ethical Hacking:**

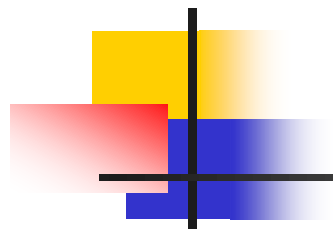
- Discovering vulnerabilities.
- Testing protection systems.
- Assessing security levels.
- Improving security policies.



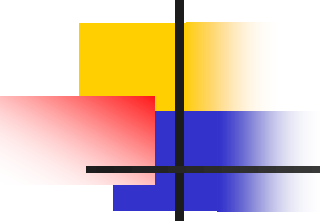
Digital Literacy

❖ **Digital Literacy Skills:**

- Critical analysis of content.
- Objective evaluation of sources.
- Deep understanding of algorithms.
- Distinguishing between facts and opinions.
- Responsible participation in the digital space.



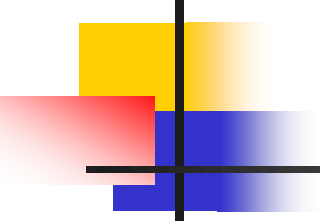
Solutions and The Way Forward



Privacy Protection Tools

❖ **Individual Protection Tools:**

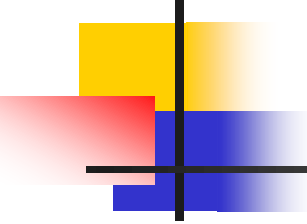
- Password Managers.
- Antivirus Software.
- VPNs.
- Two-Factor Authentication (2FA).
- Privacy-focused Browsers.



Combating Misinformation

❖ **Counter-Strategies:**

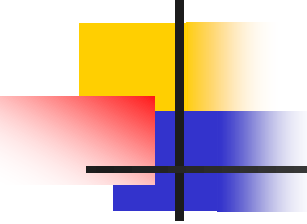
- Verifying sources.
- Analyzing context.
- Researching multiple sources.
- Using verification tools.
- Developing critical thinking



Shared Responsibility Framework

- ❖ **Stakeholder Responsibilities:**

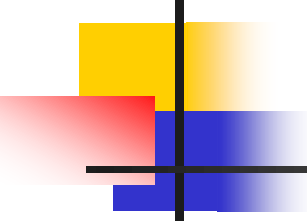
- Developers: Designing ethical systems.
- Governments: Enacting fair legislation.
- Users: Practicing responsible use



Regulatory and Governmental Solutions

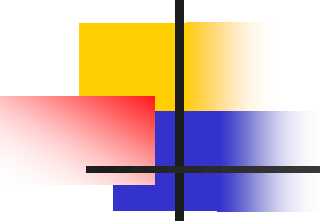
- ❖ **Regulatory Solutions:**

- Information security policies.
- Security awareness programs.
- Risk management systems.
- Cybersecurity auditing



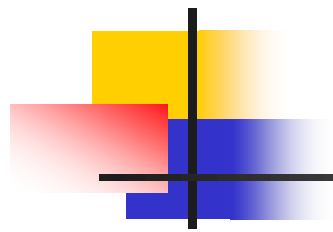
Recommendations

- Develop comprehensive legislative frameworks.
- Enhance algorithmic transparency.
- Invest in digital education.
- Strengthen international cooperation.
- Build a culture of shared responsibility.

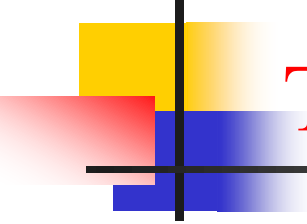


Conclusion

- ✓ Digital ethics provides an essential framework for addressing challenges arising from rapid technological evolution.
- ✓ It requires shared responsibility among individuals, organizations, and governments.
- ✓ It ensures technology is used for benefit while respecting human rights and societal values.

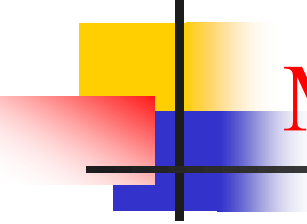


Review and Assessment



True/False Questions

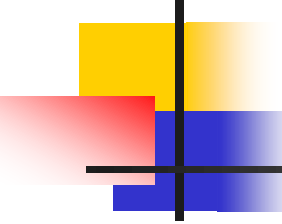
1. Digital ethics is defined solely as the study of cybercrime and fraud. F
2. Identity theft can lead to the destruction of a victim's credit rating. T
3. The "Fair Use" principle does not apply to educational purposes or F
4. The Digital Divide concerns only access to infrastructure and technology. F
5. The goal of ethical hacking is to discover vulnerabilities to improve a system's security. T
6. Protecting digital privacy is solely the responsibility of governments, not individuals. F



Multiple Choice Questions

- **Which of the following is a form of cybercrime?**
 - a) Photography.
 - b) **Phishing Attacks.**
 - c) Reading online.
 - d) Using email.

- **Which of the following scenarios often falls under the "Fair Use" principle for intellectual property?**
 - a) Selling copies of a newly released movie.
 - b) **Using short excerpts for purposes of criticism or review.**
 - c) Copying an entire commercial software program.
 - d) Illegally downloading music.

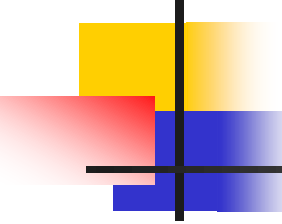


Multiple Choice Questions

- **Which of the following is NOT a level of the Digital Divide as mentioned?**
a) Access. b) Skills. c) **Entertainment.** d) Content.

- **What is one advantage of Open Educational Resources (OER)?**
a) They are expensive. b) They cannot be modified.
c) **Free Access.** d) They are developed secretly.

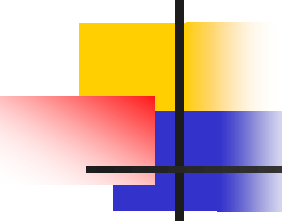
- **What is one proposed solution for combating misinformation?**
a) Believing everything you see online. b) Using a single source of information.
c) **Verifying sources.** d) Avoiding the internet.



Multiple Choice Questions

- **Which of the following is a contemporary challenge for intellectual property ?**
 - a) Lack of interest in digital content
 - b) **Difficulties in global enforcement**
 - c) Overly simple copyright laws
 - d) The high cost of obtaining patents

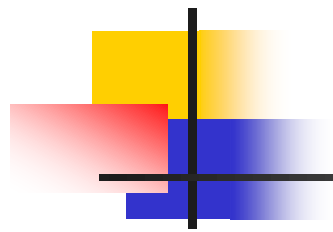
- **What is a direct impact of misinformation?**
 - a) Improved public health outcomes
 - b) Strengthening of institutional trust
 - c) **Manipulation of public opinion**
 - d) Reduction in social conflicts



Multiple Choice Questions

- According to the "Shared Responsibility Framework," what is the primary responsibility of developers?
 - a) Enacting fair legislation
 - b) Practicing responsible use
 - c) **Designing ethical systems**
 - d) Funding cybersecurity audits

- Which tool is specifically mentioned for individual privacy protection?
 - a) Ad blockers
 - b) **Password Managers**
 - c) Firewalls
 - d) Pop-up blockers



Thank You for Your Attention

Questions?